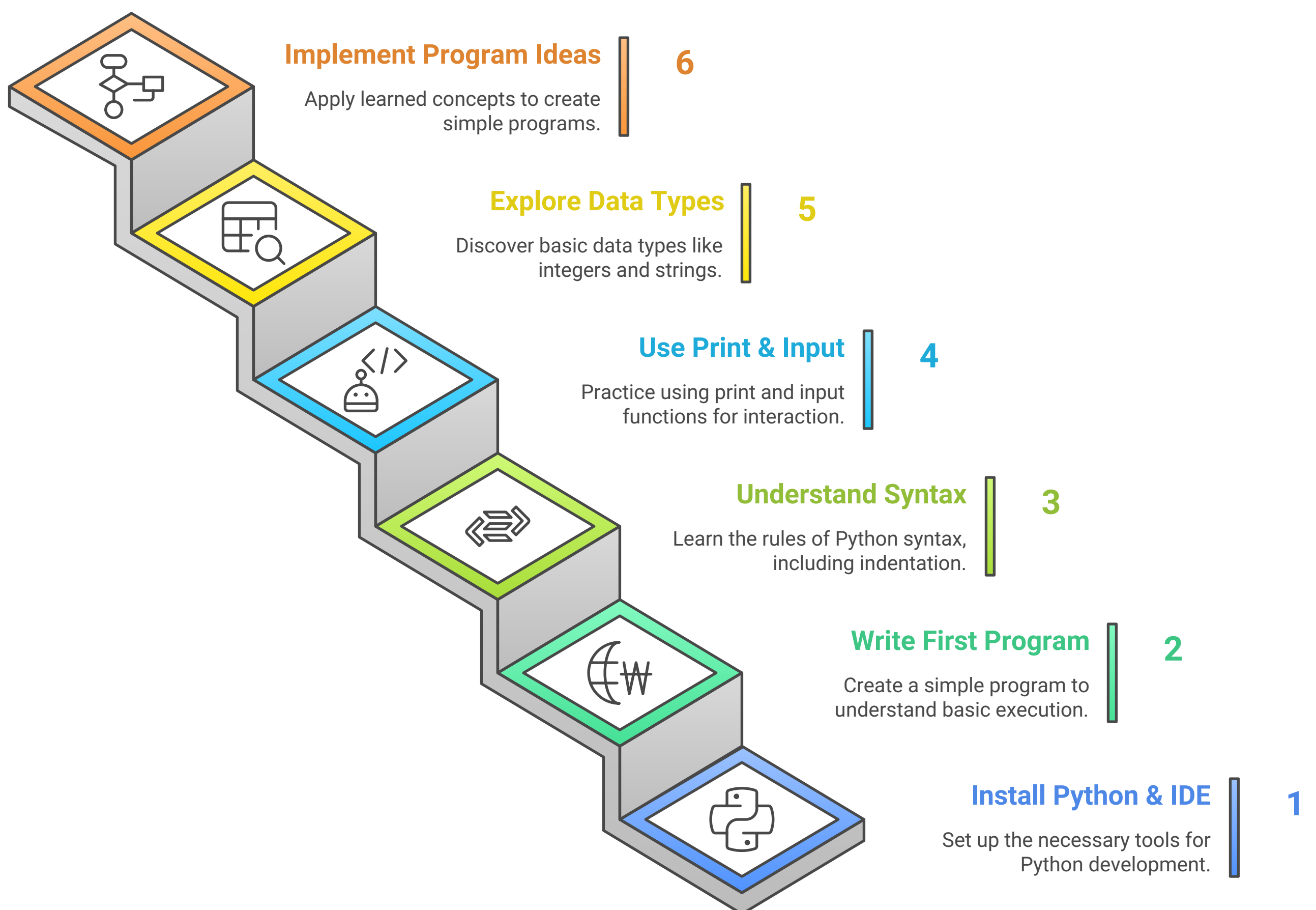




Day 1: Python Setup & Basic Syntax

This document provides a comprehensive guide to setting up Python and understanding its basic syntax. It covers installing Python and an IDE, writing your first program, understanding syntax rules like indentation and comments, using print and input functions, exploring basic data types, and implementing simple program ideas. A summary diagram is also included to consolidate the key concepts.

Mastering Python Basics



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1. Installing Python & IDE

To begin your Python journey, you need to install Python and an Integrated Development Environment (IDE).

- **Install Python:**
 - Go to [python.org][<https://www.python.org>] and download the latest version of Python suitable for your operating system.

- Follow the installation instructions provided on the website. Ensure you select the option to add Python to your system's PATH environment variable during installation. This allows you to run Python from the command line.

- **Install VS Code (recommended IDE):**

- Download Visual Studio Code from [here](https://code.visualstudio.com/).
- Follow the installation instructions for your operating system. VS Code is a lightweight but powerful source code editor that supports Python development with the help of extensions.

- **Verify Python Installation:**

- Open your command prompt or terminal.
- Type the following command and press Enter:

```
python --version
```

- This should display the version of Python you have installed, confirming that Python is correctly installed and accessible from your command line.

- **Install Python Extension in VS Code:**

- Open VS Code.
- Go to the Extensions view by clicking on the Extensions icon in the Activity Bar on the side of the window (or press Ctrl+Shift+X).
- Search for "Python" in the Extensions Marketplace.
- Install the Python extension provided by Microsoft. This extension provides rich support for Python, including syntax highlighting, IntelliSense, linting, debugging, and more.

2. Your First Python Program

Now that you have Python and VS Code set up, let's write your first Python program.

- Open VS Code.
- Create a new file by going to File > New File (or pressing Ctrl+N).
- Save the file as hello.py.
- Type the following code into the file:

```
print("Hello, Divya!")
```

- To run the program:

- Open the terminal in VS Code by going to View > Terminal (or pressing Ctrl+`).
- In the terminal, navigate to the directory where you saved hello.py using the cd command.
- Type the following command and press Enter:

```
python hello.py
```

- You should see the output "Hello, Divya!" printed in the terminal.

3. Basic Syntax Rules

Understanding Python's syntax is crucial for writing correct and readable code.

- **Indentation:**

- Python uses indentation [spaces or tabs] to define blocks of code, such as those within loops, functions, and conditional statements.
- Consistent indentation is essential; mixing spaces and tabs will result in errors. The recommended indentation is 4 spaces.

```
if True:
    print("This is indented") # Correct indentation
```

- **Comments:**

- Comments are used to explain code and are ignored by the Python interpreter.
- **Single-line comments:** Start with a # symbol.

```
- **Multi-line comments:** Enclosed in triple quotes (``"""`` or ```).

```python
```

```
"""
```

This is a multi-line comment.

It can span multiple lines.

```
"""
```

- **Case Sensitivity:**

- Python is case-sensitive, meaning that Variable and variable are treated as different identifiers.

```
Variable = 10
variable = 20
print(Variable) # Output: 10
print(variable) # Output: 20
```

### 4. Printing & Input

Python provides built-in functions for displaying output and taking user input.

- **print() function:**

- Used to display output to the console.



```
print("Hello, world!")
```

- **input() function:**
  - Used to take input from the user. It prompts the user with a message and returns the input as a string.

```
name = input("Enter your name: ")
print("Hello,", name)
```

## 5. Basic Data Types

Python has several built-in data types. Here are some of the most common:

Data Type	Example	Description
-----	-----	-----
int	x = 5	Integer (whole number)
float	pi = 3.14	Floating-point number (decimal number)
str	name = "Divya"	String (sequence of characters)
bool	is_ready = True	Boolean (True or False)

- **type() function:**
  - Used to determine the data type of a variable.

```
name = "Divya"
print(type(name)) # Output: <class 'str'>
```

## 6. Simple Program Ideas

Here are a couple of simple program ideas to practice what you've learned:

- **Age Calculator:**

```
birth_year = int(input("Enter your birth year: "))
age = 2025 - birth_year
print("You are", age, "years old.")
```

- **Pattern Printer:**

```
for i in range(1, 6):
 print("*" * i)
```

# Summary Diagram

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Python Setup	Syntax Basics	I/O Functions
Data Types		
- Install Python	- Indentation	- print()
- int, float, str		
- Install VS Code	- Comments	- input()
- bool		
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